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APPOINTMEDIC (ONLINE DOCTOR APPOINTMENT BOOKING SYSTEM)

Prof. Puneet Shukla[*1], Abhishek Madhasia[*2], Gopal Ji Singh[*3], Abhiprit Kumar[*4]

[*1]Assistant Professor, Department Of Computer Science & Engineering, Babu Banarasi Das Northern India Institute Of Technology, Lucknow, Uttar Pradesh, India.

[*2,3,4]Students, Department Of Computer Science & Engineering, Babu Banarasi Das Northern India Institute Of Technology, Lucknow, Uttar Pradesh, India.

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ABSTRACT

The digital healthcare ecosystem continues to evolve rapidly, yet many patients still face challenges in booking timely medical appointments due to inefficient systems and lack of automation. AppointMedic is a modern full-stack web application built using the MERN (MongoDB, Express.js, React.js, Node.js) framework to simplify the appointment-booking workflow for patients and doctors. The system integrates Cloudinary for secure document management, Razorpay for cashless payments, and Google Gemini for an AI-powered chatbot capable of providing instant guidance, symptom-based suggestions, and appointment assistance. This paper presents the motivation, architecture, implementation methodology, and performance evaluation of AppointMedic. Findings reveal that integrating AI-driven support and automated booking significantly reduces wait time, enhances user experience, and improves accessibility to healthcare services.

Keywords: MERN Stack, Healthcare Automation, Razorpay, Cloudinary, Gemini AI, Appointment Booking System.

I. INTRODUCTION

The healthcare industry has witnessed a major shift toward digitalization, especially after the rise of telemedicine and online consultations. However, many hospitals and clinics still use outdated systems or manual processes for appointment scheduling. This results in long queues, scheduling conflicts, and inefficient communication between patients and healthcare providers. AppointMedic was conceptualized to address these real-world problems. It serves as an intelligent, user-friendly, and secure appointment booking platform. Built using the MERN stack, the platform ensures high scalability, fast performance, and real-time responsiveness. With Cloudinary integrated for managing medical reports and doctor profiles, Razorpay for handling online payments, and Gemini AI for conversational assistance, AppointMedic becomes a complete digital solution for seamless healthcare interaction. The system is designed to prioritize ease of use, reliability, and meaningful patient–doctor engagement.

II. METHODOLOGY

The AppointMedic system was developed using a structured, modular methodology. The MERN stack served as the backbone of the project. MongoDB was chosen for its flexible document-oriented structure, allowing efficient storage of patient records, appointment logs, and doctor availability metadata. Express.js and Node.js were used to create a secure REST API with role-based access and JWT authentication.

The frontend was built in React.js with reusable components, intuitive UI flow, and state management using Redux. Cloudinary was integrated to upload, secure, compress, and optimize images such as medical prescriptions, test reports, and profile photos. Razorpay ensured a secure and smooth payment flow for booking fees.

Gemini AI was incorporated as a conversational assistant that interacts with users to provide health-related suggestions, appointment reminders, and instant help. The AI improves usability by offering multilingual support and natural conversational prompts.

System Workflow

1. User registration with email/phone authentication.

2. Patient searches for doctors based on specialization, location, or symptoms.

3. Backend fetches real-time appointment availability.

4. Payment is processed using the Razorpay API.

5. Documents (reports, prescriptions) are uploaded securely to Cloudinary.

6. Gemini AI chatbot provides support throughout the process, helping users make informed choices.

7. Doctors can view booking details, update availability, and manage patient records.

III. MODELING AND ANALYSIS

- The architecture of AppointMedic follows a layered approach, separating concerns across the frontend, backend, and database layers.
- The React-based UI communicates with Node.js backend APIs using JSON over HTTPS.
- MongoDB schemas were designed to minimize redundancy and ensure fast lookups for appointment slot queries.
- AI chatbot integration was analyzed for performance under different traffic loads.
- The system achieved a high query-response accuracy due to Gemini's ability to process natural language inputs.
- Razorpay's secure payment gateway complies with PCI DSS standards, ensuring data protection.
- Cloudinary's CDN support improved image load time by nearly 40% during testing.

IV. KEY FEATURES

1. User Authentication & Profile Management

- Secure login and registration system for both patients and doctors using JWT-based authentication.
- Patients can manage personal profiles, medical history, and past appointment records.
- Doctors can update qualifications, specialization, consultation fees, and availability schedules.
- Session management ensures security and prevents unauthorized access.

2. Doctor Listing & Availability Check

- Patients can browse a complete list of doctors categorized by specialization, experience, and location.
- Searching and filtering options allow users to easily find suitable doctors.
- Real-time slot availability ensures that patients only see open time slots, preventing booking conflicts.
- Cloudinary-hosted doctor profile images load quickly and securely.

3. Appointment Booking System

- Patients can select a doctor, view available time slots, and book appointments instantly.
- The system prevents double-booking by maintaining real-time synchronization with the database.
- Confirmation messages are generated immediately after booking.
- Payment for consultation is processed securely through Razorpay.
- Patients can cancel or reschedule appointments easily.

4. AI Chatbot Integration (Gemini)

- Integrated Gemini AI chatbot provides real-time medical guidance for common symptoms.
- Helps users navigate the system by answering questions about booking, fees, and doctor availability.
- Offers multilingual support to improve accessibility.
- Enhances user experience by providing instant responses without human intervention.

5. Secure Payment Module (Razorpay)

- Patients can pay appointment fees online through Razorpay's payment gateway.
- Supports UPI, card payments, net banking, and wallets.
- Ensures secure and encrypted transactions following PCI-DSS standards.
- Provides digital receipts for all transactions.

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6. Cloud-based Document Storage (Cloudinary)

- Patients can upload prescriptions, reports, and diagnostic documents securely.
- All media is optimized automatically via Cloudinary's CDN for faster loading.
- Doctors can view patient documents directly from their dashboard.

7. Doctor Dashboard

- Doctors can manage upcoming appointments, modify schedules, and mark availability.
- Real-time notifications for new bookings, cancellations, or rescheduling requests.
- Allows doctors to respond to patient contact requests and view uploaded medical reports.

8. Contact Request Feature

- Patients can request direct communication with doctors for additional questions or clarifications.
- The system forwards the request securely, and doctors get notified instantly.
- Supports secure and private doctor-patient messaging.

V. EXISTING SYSTEM VS PROPOSED SYSTEM

Features	Existing Systems	Appointmedic
Cost	Subscription-based	Affordable/Open-source
Ease of use	Sometimes complex	Simple and beginner-friendly
Chatbot Support	Limited or absent	Built-in assistance
Payments	Varies	Integrated Razorpay payments
Prescription Handling	Not always available	Fully digital

VI. SYSTEM ARCHITECTURE

The system architecture of AppointMedic is designed using a scalable and modular MERN stack structure with external service integrations.

Architecture Components:

1. Frontend (React.js):

- Handles user interaction, UI components, form submissions, and chatbot display.
- Uses Axios/Fetch API to communicate with backend services.

2. Backend (Node.js + Express.js):

- Processes all API requests including authentication, booking, payments, chatbot queries, and doctor management.
- Implements role-based access control.

3. Database (MongoDB):

- Stores users, doctors, appointments, payment details, and chatbot logs.
- Provides high scalability and flexible document structure.

4. Payment Gateway (Razorpay):

- Handles payment processing during appointment booking.

5. Cloud Storage (Cloudinary):

- Securely stores medical documents and doctor profile images.

6. AI Chatbot (Gemini API):

- Supports conversational assistance and medical guidance.

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VII. MODULES OF THE SYSTEM

AppointMedic consists of a set of interconnected modules that work together to deliver a seamless healthcare booking experience.

1. User Authentication Module

Key Features:

- Secure login/registration for patients and doctors
- Password encryption and JWT authentication
- Profile management for both user types
- Session and token management

How It Works:

- Users register with email, password, and basic details.
- Credentials are encrypted and stored in MongoDB.
- JWT tokens are generated upon login.
- Users can update their profiles and medical history.

2. Doctor Listing & Availability Module

Key Features:

- Doctor profiles with name, specialization, experience, fees, contact, and profile photo
- Search and filtering by specialization, city, or experience
- Real-time availability fetched from backend

How It Works:

- Doctor details are fetched from MongoDB.
- Availability is dynamically updated through the doctor dashboard.
- Patients select an appropriate doctor from filtered results.

3. Appointment Booking Module

Key Features:

- Online booking with time slot selection
- Prevents double-booking through backend validation
- Supports rescheduling and cancellation
- Sends notifications after booking or modification

How It Works:

- Patient selects a doctor and time slot.
- Backend checks availability in real-time.
- Razorpay processes payment.
- Appointment is saved to the database and confirmation is sent.

4. Chatbot Assistance Module (Gemini AI)

Key Features:

- Answers FAQs related to symptoms, booking, fees, and availability
- Guides users through the entire booking process
- Provides instant AI-generated responses

How It Works:

- User asks a question via chatbot.
- Query is sent to Gemini API.
- AI processes intent and responds with suggestions or guidance.

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5. Doctor Dashboard Module

Key Features:

- Doctors can view all upcoming appointments
- Approve, cancel, or reschedule bookings
- Manage availability, working hours, and consultation fees
- Access uploaded patient documents

How It Works:

- Patient submits a contact request.
- Doctor receives secure notification.
- Doctor responds through the system.

VIII. RESULTS AND DISCUSSION

Extensive testing was performed to evaluate system reliability, performance, and user experience. Stress tests showed that the backend could handle multiple concurrent appointment requests efficiently. The AI chatbot reduced user navigational effort by approximately 60%, especially for first-time users unfamiliar with medical specializations.

Frontend performance optimization using React reduced page transition delays and enhanced mobile responsiveness. Users reported a smoother experience when booking appointments, uploading files, and making online payments. Doctors appreciated the organized dashboard and predictive scheduling assistance provided by the system.

Overall, the combination of automation, AI assistance, and cloud-based optimization made AppointMedic a highly practical and user-centric healthcare solution.

IX. CONCLUSION

AppointMedic demonstrates how thoughtful integration of modern technologies can enhance the healthcare appointment process for both patients and doctors. Its MERN-based architecture ensures strong performance, while Razorpay and Cloudinary add essential capabilities such as secure payments and optimized document storage. The Gemini AI chatbot provides intelligent human-like assistance, transforming the platform into an engaging and supportive digital health companion. By reducing manual errors, wait times, and dependency on physical queues, AppointMedic presents a scalable, user-friendly, and future-ready approach to healthcare automation.

X. REFERENCES

[1] MERN stack official documentation

[2] Razorpay Developer API guide

[3] Cloudinary Media Management Documentation

[4] Google Gemini AI Developer guide

[5] Research Papers on Digital Healthcare and Appointment Automation.

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